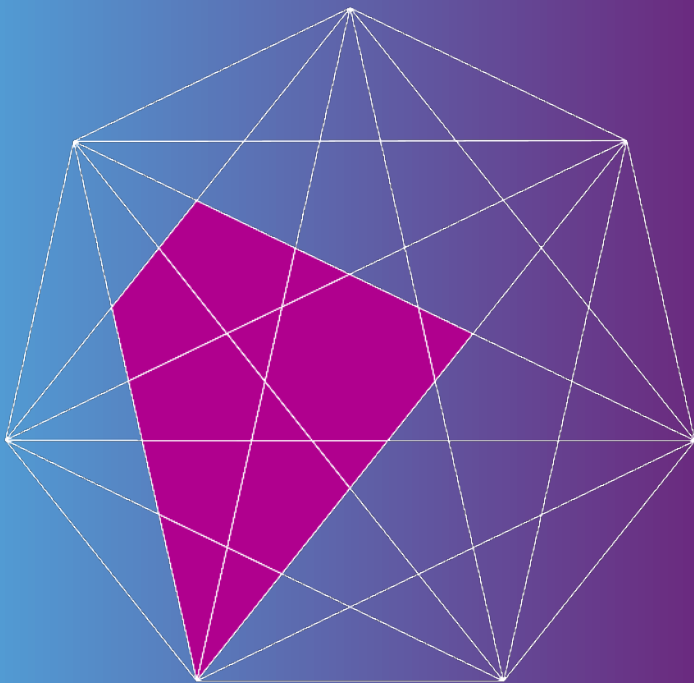




VNS Therapy™ Surgical Procedure for New Implantations



INTENDED USE / INDICATIONS

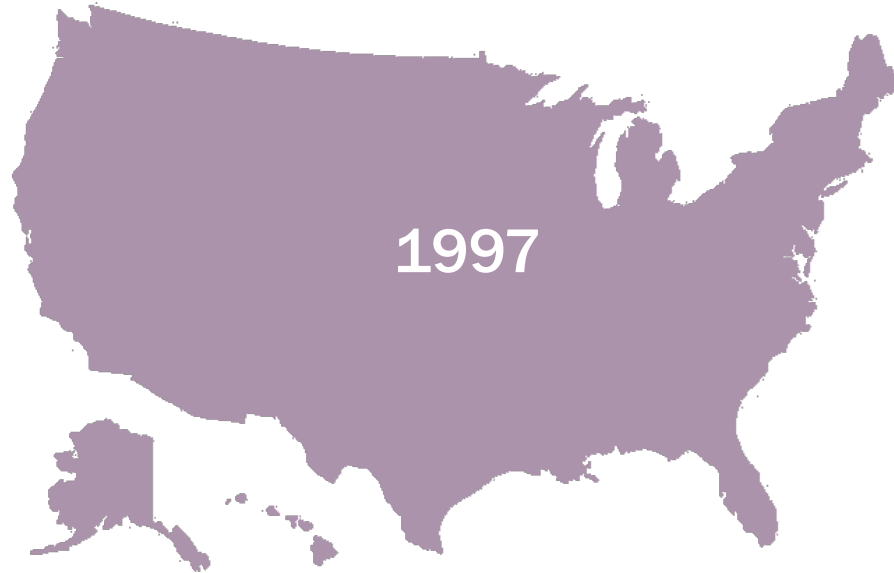
Epilepsy (US)—The VNS Therapy System is indicated for use as an adjunctive therapy in reducing the frequency of seizures in patients 4 years of age and older with partial onset seizures that are refractory to antiepileptic medications.

LivaNova
Epilepsy



An Introduction to VNS Therapy™

VNS Therapy™ has been FDA approved for clinical use in the USA since...



US indication for use¹

The VNS Therapy System is indicated for use as an adjunctive therapy in reducing the frequency of seizures in patients 4 years of age and older with partial onset seizures that are refractory to antiepileptic medications.

Local labeling should be referenced for specific indications for use by geography. 1. VNS Therapy™ System Epilepsy Physician's Manual, May 2020, LivaNova USA, Inc. Houston, TX.

IM-7600387-EPI

25 Years of Proven Experience

>130,000



patients treated¹

>40,000



children treated¹



75%

reimplantation rate^{1*}



Prescribed
neuromodulation
device for DRE¹

4+ years

Only DRE neurostimulation
therapy indicated to treat children
as young as 4 years old²

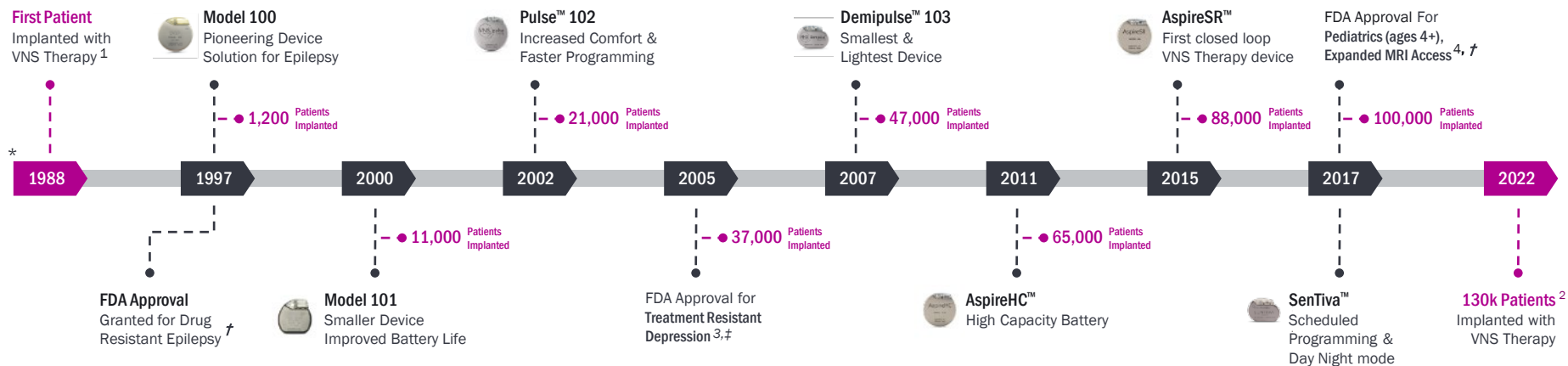
*Of those eligible for reimplantation

1. Data on File, LivaNova, Inc. Houston, TX. As of April 2020.

2. VNS Therapy™ System Epilepsy Physician's Manual, May 2020, LivaNova USA, Inc. Houston, TX.

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A Brief History of VNS Therapy™ Innovation



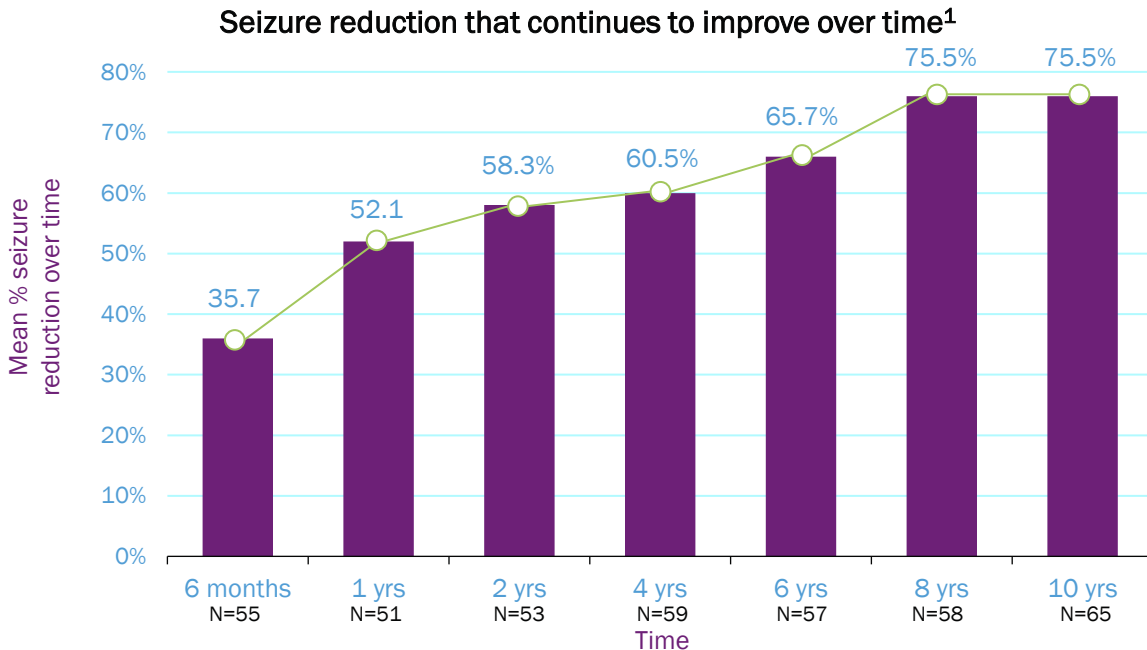
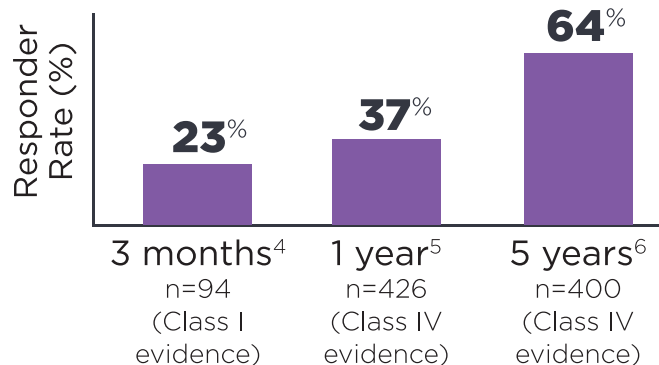
1. First Patient Implanted Data: Penry, JK et al. *Epilepsia*. 1990;31:S40-S43 2. Patient implant data on file. LivaNova USA, Inc. 3. [Accessdata.fda.gov](https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpma/pma.cfm?id=P970003S050), 2005, www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpma/pma.cfm?id=P970003S050. 4. FDA. 2017. Available from: <https://www.fda.gov/downloads/MedicalDevices/ProductsandMedicalProcedures/DeviceApprovalsandClearances/PMAApprovals/UCM565765.pdf>.

† **Epilepsy (US)**—The VNS Therapy System is indicated for use as an adjunctive therapy in reducing the frequency of seizures in patients 4 years of age and older with partial onset seizures that are refractory to antiepileptic medications.

‡ **Treatment Resistant Depression (US)**—The VNS Therapy System is indicated for the adjunctive long-term treatment of chronic or recurrent depression for patients 18 years of age or older who are experiencing a major depressive episode and have not had an adequate response to four or more adequate antidepressant treatments.

*Dates correspond with commercial launch of each device

VNS Therapy™ may continue to reduce seizure frequency with increasing effect over time



Seizure frequency significantly reduced from baseline at each recorded interval over 10 years ($p < 0.001$) in patients with partial and generalized seizures¹

*Data includes both partial and generalized seizures, however VNS Therapy is only indicated for partial onset seizures.

1. Elliott RE, et al. *Epilepsy Behav.* 2011;20:478-83. 4. Handforth A, et al. *Neurology.* 1998; 51:48-55.; 5. Morris GL, et al. *Neurology.* 1999; 53:1731-5. 6. Elliott RE et al. *Epilepsy Behav.* 2011; 20:57-63.

VNS Therapy™ may reduce the economic burden of untreated DRE

VNS Therapy is associated with decreased resource utilization and epilepsy-related clinical events¹

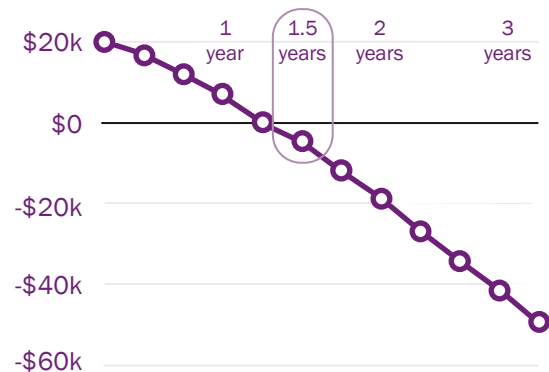
The following clinical events were reduced¹

Average follow-up 30.4 months

N = 1655

Event	Percent Reduction
ER visits	39%
Seizure-related hospitalizations	47%
Seizure-related outpatient visits	17%

Total healthcare cost savings (USD) begin at 1.5 years¹



Negative net costs indicate lower costs in the post-VNS Therapy period relative to the mean quarterly cost in the pre-VNS Therapy period (N=1,655)

DRE, drug-resistant epilepsy. USD, United States dollars.
1. Helmers S, et al. *Epilepsy Behav.* 2011;22:370–75.

VNS Therapy™ Safety Profile - Stimulation

VNS Therapy™ is not associated with systemic CNS-related side effects and does not add to the burden imposed by ASMs

Stimulation related side effects associated with VNS Therapy™

- Usually occur during stimulation and generally diminish over time^{1,2}
- May be diminished or eliminated by the adjustment of parameter settings³ and may be controlled by use of the magnet⁴
- Incidence of adverse events following stimulation (>5%) were voice alteration, increased coughing, pharyngitis, paresthesia, dyspnea, dyspepsia, and nausea.

Side Effect ²	Year 1	Year 2	Year 3
Hoarseness	29%	19%	2%
Paresthesia	12%	4%	0%
Cough	8%	6%	2%
Shortness of Breath	8%	3%	3%



1. Ben-Menachem E. *J Clin Neurophysiol.* 2001;18(5):415-418. 2. Morris GL, et al. *Neurology.* 1999;53(7):1731-1735. 3. Heck C, et al. *Neurology.* 2002;59(6 Suppl 4):S31-37. 4. VNS Therapy™ System Epilepsy Physician's Manual. Houston, TX: LivaNova USA Inc.; 2020.
CNS= Central Nervous System. ASM = Anti-seizure Medication

VNS Therapy™ Safety Profile - Surgical Procedure

Surgical side effects associated with VNS Therapy™ in Adults

- Infection is the most common side effect associated with the surgical procedure¹
- Surgical side effects occur at a rate of 4.1% in adult patients aged ≥ 18 years¹, including infection, nerve paralysis, hypesthesia, facial paresis, left vocal cord paralysis, left facial paralysis, left hemidiaphragm paralysis, left recurrent laryngeal nerve injury, urinary retention, and low-grade fever

Surgical side effects associated with VNS Therapy™ in Pediatric Patients

- Infection is the most common side effect associated with the surgical procedure¹
- Infection due to surgery is 6.4% in pediatric patients aged 4-11 years¹ with 1.1% of cases resulting in extrusion of lead
- Infection due to surgery is 3.4% in patients aged 12-21 years¹ with 0.26% of cases resulting in extrusion of lead
- Because of greater risk of infection, pediatric patients should be monitored for site infection and avoid manipulation of the surgical site¹



VNS Therapy™ – System Components

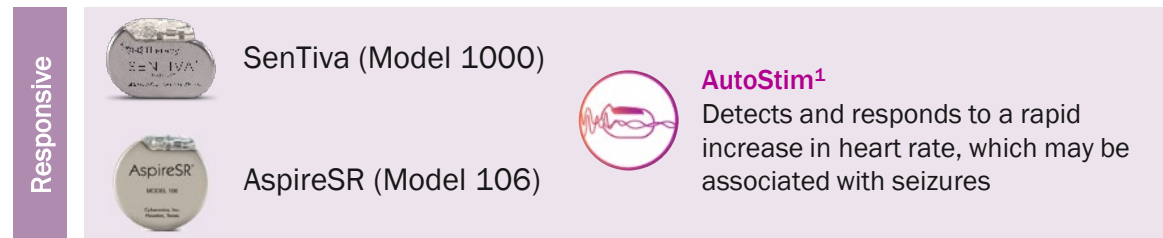
VNS Therapy™ System components

Implantables		Surgical tools
 VNS Therapy™ generator	 Bipolar electrical lead	 Tunneler
Non-implantables		 Accessory pack
 Programmer and wand	 Patient Kit (including magnet)	

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The generator is the main implantable component of the VNS Therapy™ System

- The primary implantable component of the system is the VNS Therapy generator, which may be referred to as a ‘stimulator’²
- The generator is computer controlled and battery powered²



Parameter settings and battery life

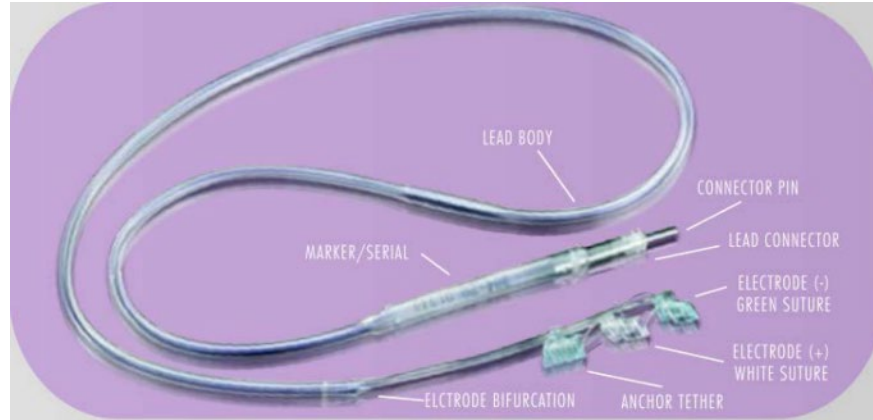
- Table shows the impact of the AutoStim feature on the longevity of battery life. These calculations are based on typical lead impedance (3 kOhms) with Normal Mode settings: 2 mA output current, 20 Hz signal frequency, 250 µsec pulse width and 10% duty cycle (30 seconds ON and 5 minutes OFF)
- Certain combinations of parameter settings may decrease battery life if those parameter settings result in an increased rate of stimulation.

Estimated Battery Longevity		
	Model 1000	
AutoStim Feature	Expected Life (yrs)	
Tachycardia Detection/AutoStim “OFF”	11.5	
	AutoStim ON-Time	
Tachycardia Detection/AutoStim “ON”	30 sec	60 sec
AutoStims Per Hour	Expected Life (yrs)	
1	7.6	7.5
7	7.0	6.1
15	6.4	4.9

1. VNS Therapy™ System Epilepsy Physician’s Manual, LivaNovaUSA, Inc. Houston, TX. 2. Patient’s Guide for Epilepsy, LivaNova USA, Inc. Houston, TX. Available at: <https://vnstherapy.com/patient-resources>.

The lead carries stimulation from the generator to the vagus nerve

PerenniaFLEX™, Model 304



3.5x more durable than previous generation lead (Model 302)

The programming system¹

- **Wireless** connectivity between programmer and wand
- Programmer and wand will communicate at a distance of up to **3 meters (~10ft)**

The latest generation of the VNS Therapy™ System also offers:*

Scheduled Programming

- Automatic increase of stimulation therapy parameters without office visit

Day-Night Programming

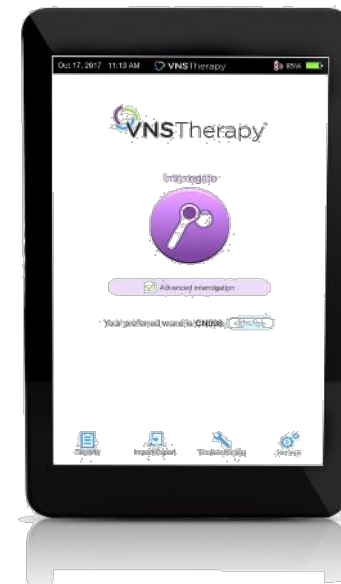
- An optional feature that allows the generator to deliver 2 independent sets of therapy parameters at different times during a 24-hour period

*For SenTiva (Model 1000)

1. VNS Therapy™ System Epilepsy Physician's Manual, May 2020, LivaNova USA, Inc. Houston, TX.



Programming wand
Model 2000



Programmer (tablet and software)
Model 3000

The model 402 tunneler is used during surgery to guide the lead¹

- The tunneler is a **single-use** surgical tool
- It is used for subcutaneous tunneling of lead from neck to generator pocket
- May be manually shaped (≤ 25 degrees) to help direct it through the body
- **Compatible with ALL VNS Therapy System leads**



1. VNS Therapy™ System Epilepsy Physician's Manual, May 2020, LivaNova USA, Inc. Houston, TX.



New Implant Procedure

Procedure Overview

1 Expose carotid sheath and left vagus nerve

2 Create a pocket in the chest for the generator

3 Choose the correct lead size

4 Tunnel the lead

5 Attach the electrodes

6 Form the strain relief bend and loop

7 Connect the lead to the generator

8 Perform the pre-insertion system diagnostics test

9 Place the generator in the chest pocket

10 Perform the second system diagnostics test

11 Interrogate the generator

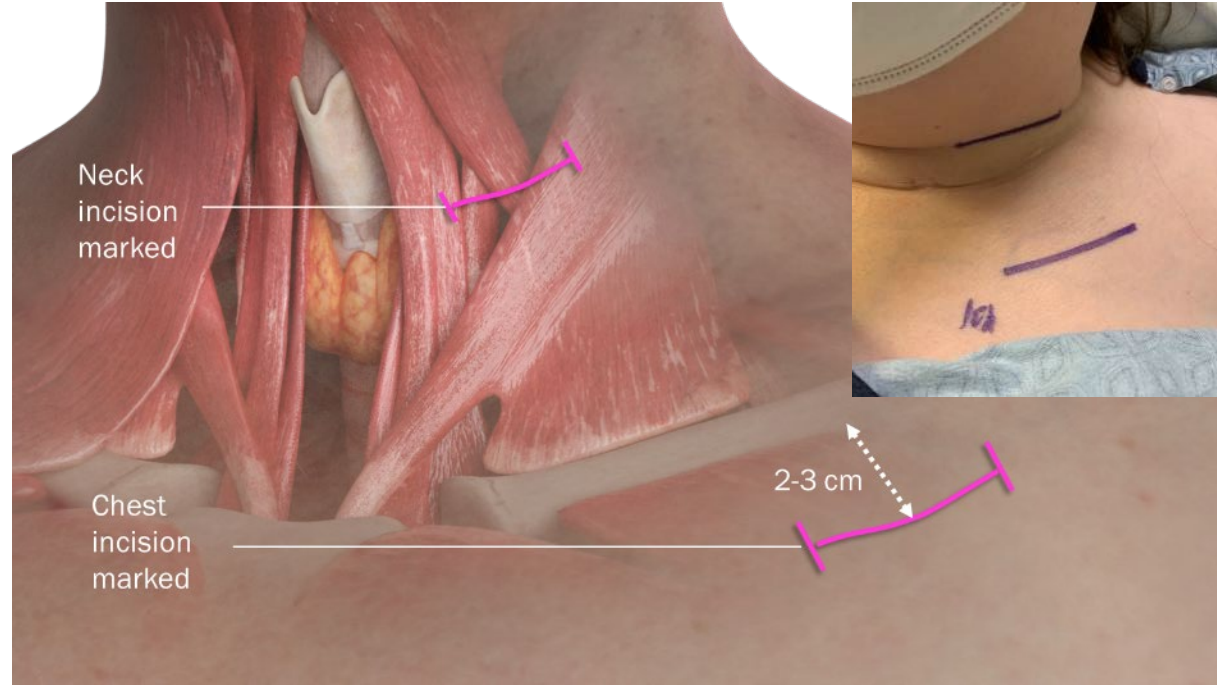
12 Irrigate and close the incision

Preparing the patient

Suggested placement for the lead is the area **half-way between the clavicle and the mastoid process**

Generator is usually implanted **just below the clavicle** in a subcutaneous pocket in the left upper chest

- May also be placed along the axillary border



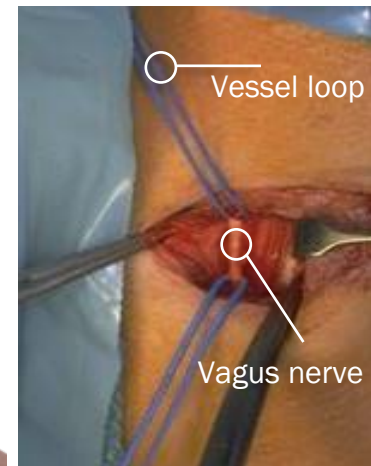
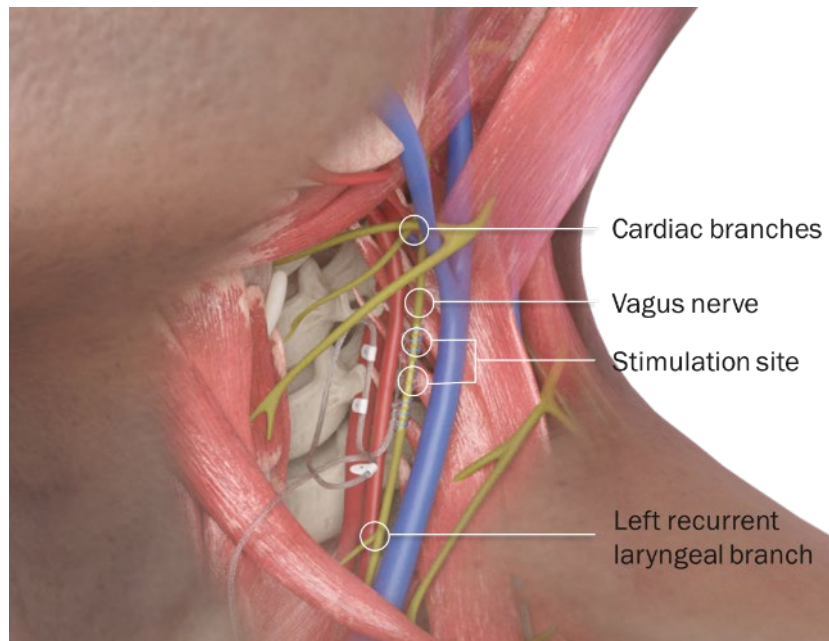
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Step 1: Expose the carotid sheath and left vagus nerve

Expose ≥ 3 cm of nerve

- May use soft rubber vessel loops to raise or lift the nerve as necessary
- Ensure that the nerve is handled with care

Avoid excessive handling of nerve to prevent injury and preserve branches off the vagus nerve



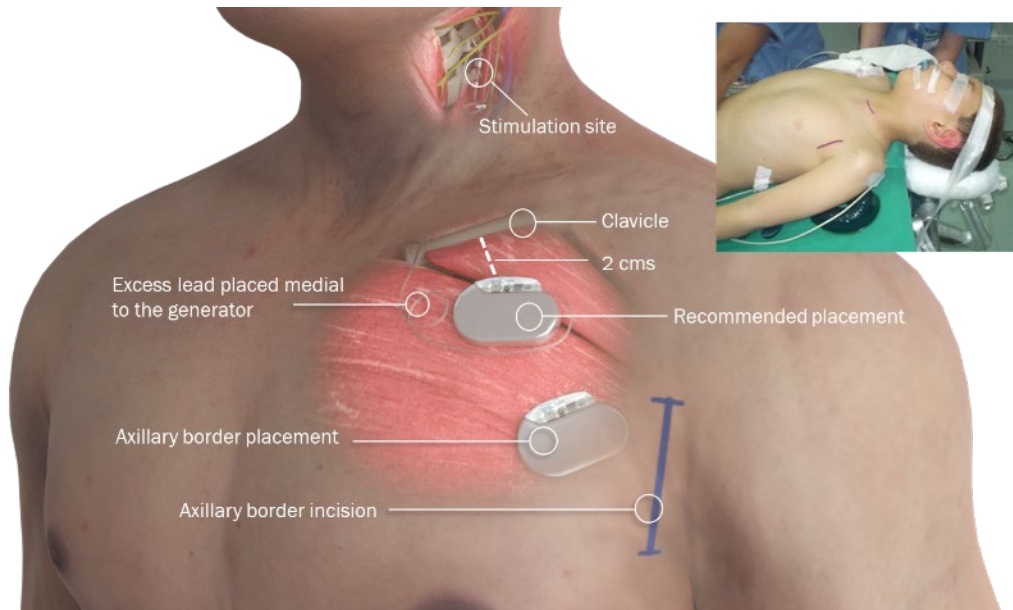
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Step 2: Create a pocket in the chest for the generator

Create a subcutaneous pocket in either the left upper chest (below clavicle) or along the axillary border for the generator and excess lead

Successful communication is most likely if the surface of the programming wand head is within **1 inch** of the generator

- It is not recommended to implant generator below muscle. Doing so may contribute to communication difficulties once implanted



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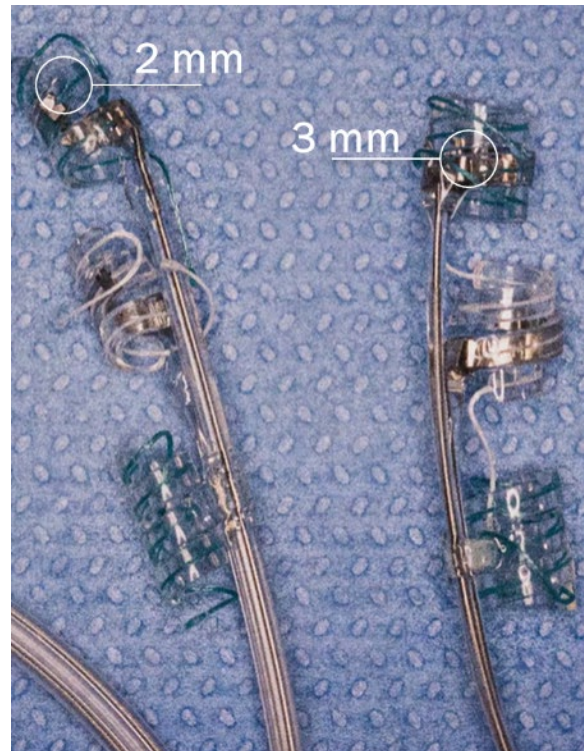
Step 3: Choose the appropriate lead size

Choose the appropriately-sized lead (2.0mm or 3.0mm electrode inner diameter)

- Assess size of vagus nerve
- Lead should fit snugly without constricting the nerve
- 2.0mm should accommodate most nerves

DO NOT soak the lead or generator in saline or similar solution before implanting

- May cause the insulated portions of the connector pin to swell and become difficult to insert into the generator



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Step 4: Tunnel the lead

Preparing the lead and tunneler

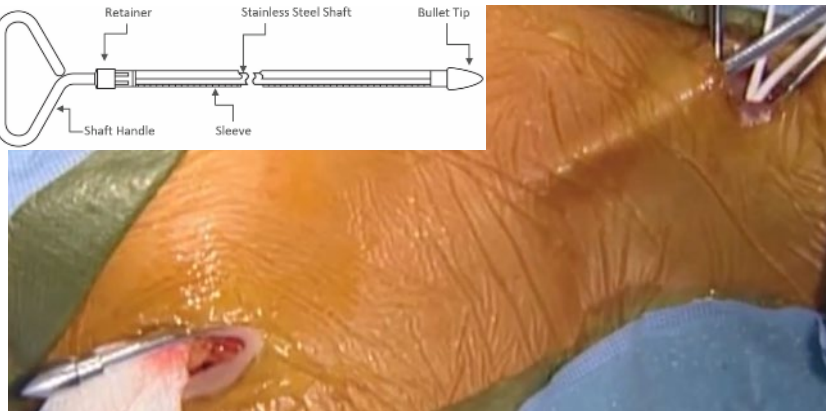
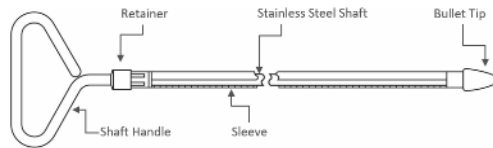
- Place smaller tunneling sleeve over the tunneling shaft and hand thread the bullet tip at the end of the shaft

Passing the tunneler

- Place the bullet tip end of the tunneler at the neck incision, and tunnel subcutaneously toward the chest incision, exerting force on the handle end and directing the tunneler as necessary

Pass the lead

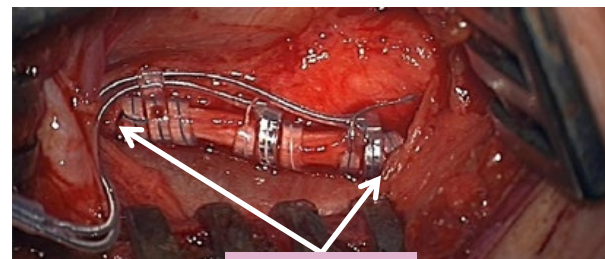
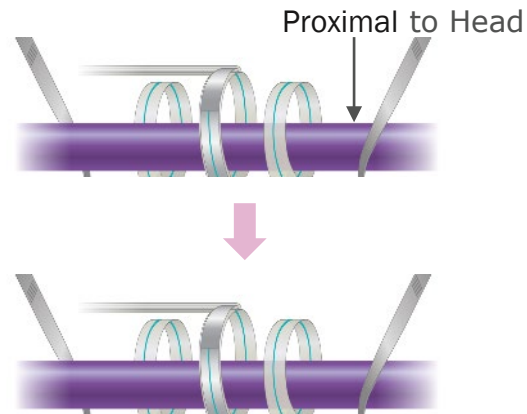
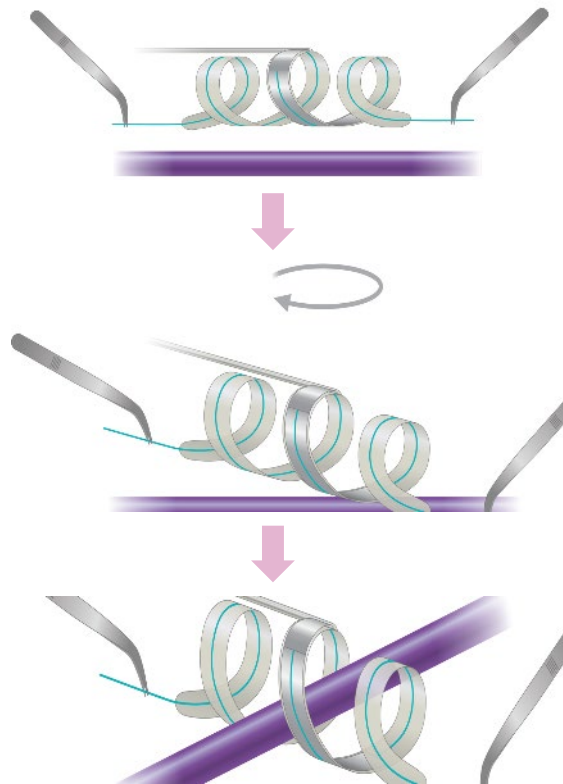
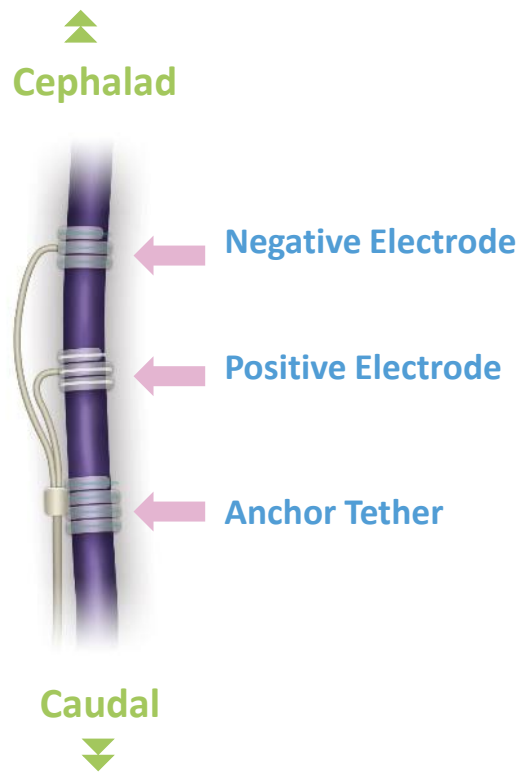
- With the sleeve in place between the two incisions, carefully insert the lead connector pin **inside the end of the sleeve** at the neck incision
- Carefully pull the sleeve along with the lead connector pin from the chest end until the lead connector pin **completely exits** the chest incision



Lead pin inserted into tunneler sleeve at neck incision

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Step 5: Attach the electrodes

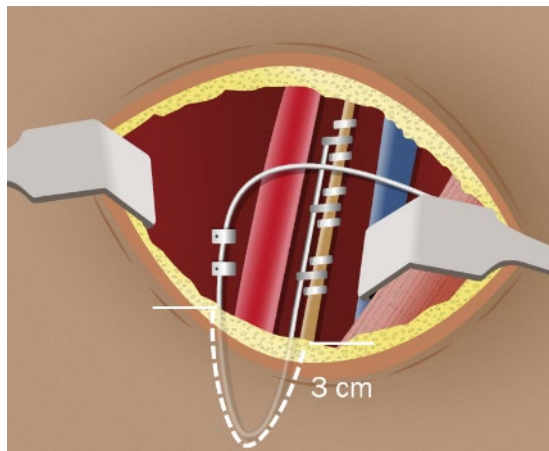


Secured electrodes and wire in pocket

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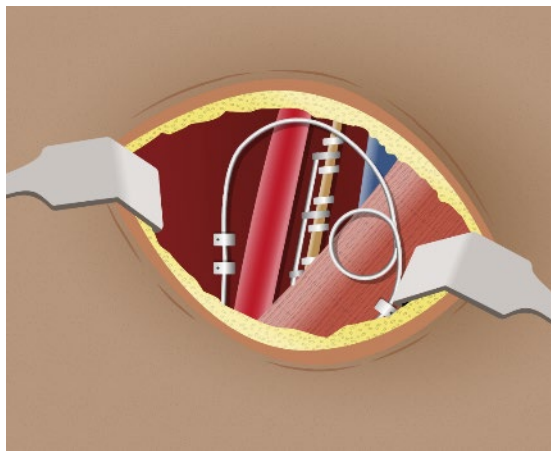
Step 6: Form the strain relief bend and loop

Strain Relief Bend



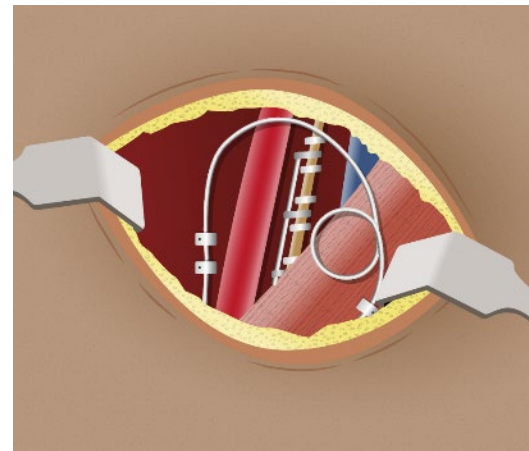
1. Run **at least 1 cm** of lead parallel to the vagus nerve
2. Place **3 cm strain relief** bend

Strain Relief Loop



1. Place a **strain relief loop** as the lead comes over the sternocleidomastoid muscle
2. Loosely **attach it to fascia** with a tie-down before routing the lead over the clavicle

Use of Tie-Downs



1. Secure strain relief with silastic tie-downs provided within lead package

Step 7: Connect the lead to the generator

1. Look inside the generator receptacle(s) and **verify that the path is clear**. If necessary, reverse set screw(s) to clear the path. If the set screw must be reversed, **avoid backing it out completely** – use no more than two counterclockwise turns
2. Insert the hex screwdriver into the setscrew plug to **relieve back pressure** during lead insertion

Note: Do not use electrosurgical equipment after the new generator has been introduced into the sterile field. Exposure to this equipment may damage the generator.



Step 7: Connect the lead to the generator

3. Insert the connector boot fully into the header of the generator
4. Verify that the connector pin is fully inserted. The pin should be visible in the area at the back end of the setscrew connector block
5. Fully insert the hex screwdriver, and press down while tightening the set screw until you hear a click
6. Pull back gently on the connector boot to ensure that the lead is secure

Note: Failure to make a proper connection and to completely tighten the setscrew can result in high lead impedance and/or erratic stimulation at varying intensities



1. VNS Therapy™ System Epilepsy Physician's Manual, May 2020, LivaNova USA, Inc. Houston, TX.

Step 8: Perform the pre-insertion system diagnostics test

The **programming wand** should be placed into a sterile laser arm bag, camera bag, or equivalent for entry into the sterile field

Handheld computer can remain outside the sterile field or can be passed through laser arm bag with programming wand if the surgeon will be performing the diagnostic tests

Note: Sterile laser arm bag not provided by LivaNova.



Step 8: Perform the pre-insertion system diagnostics test

Select Perform Diagnostics, then System Test

End session Interrogate

Patient ID: TOD
SenTiva M1000 S/N: 93657
Implant Date: Dec 1, 2017

Diagnostics
Measured on: Jun 4, 2018 9:10 PM
Output Current: **1.25 mA**
Lead Impedance: **OK**
Generator Battery: **OK**

Events
Magnet stims per day (since last visit): **0.00**
AutoStim events per day (since last visit): **0.00**

Perform Diagnostics View all

Parameters
Stimulation

	NORMAL	AUTOSTIM	MAGNET
Output	1.25 mA	1.375 mA	1.5 mA

Tachycardia Detection: **ENABLED**
Autostim Threshold: **70 %**

Edit Parameters

Summary Parameters Diagnostics Events History More

End session Interrogate

Patient ID: TOD
SenTiva M1000 S/N: 93657
Implant Date: Dec 1, 2017

DIAGNOSTICS

Diagnostic Tests

System Magnet AutoStim

Checks if the System is working correctly
Checks if Magnet Stimulation is working correctly
Checks if Automatic Stimulation is working correctly

Test Test Test

Diagnostics Test History

TEST	DATE	LEAD IMPEDANCE	BATTERY STATUS
✓ System	Jun 4, 2018	✓ 4254 Ohms	✓ OK
✓ System	May 31, 2018	✓ 4254 Ohms	✓ OK
✓ System	May 31, 2018	✓ 4254 Ohms	✓ OK
✓ System	May 31, 2018	✓ 4254 Ohms	✓ OK
✓ System	May 31, 2018	✓ 4254 Ohms	✓ OK
✓ System	May 31, 2018	✓ 4254 Ohms	✓ OK

Summary Parameters Diagnostics Events History More

Diagnostics Test Results
Jul 24, 2018, 6:28 PM

System Diagnostics Test Results

Lead Impedance **>= 9000 Ohms** **HIGH**

Generator Battery 75% - 100% **OK**

Output Current **2 mA** **LOW**

Done

Impedance Value	Lead Impedance
≤ 600 Ohms	Impedance LOW
600 to ≤ 5,300 Ohms	Impedance OK
5,300 to 10,000 Ohms	Impedance HIGH
≥ 10,000 Ohms	Impedance HIGH

Troubleshooting High/Low Lead Impedance

1. **Remove lead pin**, re-insert lead pin (ensure using correct technique), and run System Diagnostic test again. If nerve site is dry, irrigate nerve and remove pooled fluid
 - If impedance is “OK”, proceed with implant
 - If impedance is “High” or “Low”, continue to step 2
2. **Check electrodes** are properly and completely wrapped around the nerve
 - If no visible abnormalities exist, continue to step 3
3. **Perform Generator Diagnostic Test** with test resistor
 - If impedance is “OK”, continue to step 4
 - If impedance is “High” or “Low”, call Customer Quality
4. **Reinsert lead pin** and perform System Diagnostics again
 - If impedance is “OK”, proceed with implant
 - If impedance is “High” or “Low”, call Customer Quality

Steps 9 – 11: Place generator in the chest pocket, perform the second system diagnostics test, and interrogate the generator

- Place generator in chest pocket
- **Coil the remaining lead** and place to the side of the generator
- Logo may face up or down
- **Secure generator** by placing suture through suture hole and attaching to fascia with non-absorbable suture

- ✓ After placing the generator inside the pocket and before the skin is closed, **perform a 2nd System Diagnostic test**
- ✓ **Interrogate** to verify all output currents are programmed to 0.0 mA



Note: Do not place lead slack under the generator as doing so could result in insulation failure and system malfunction

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Generators Capable of Tachycardia Detection

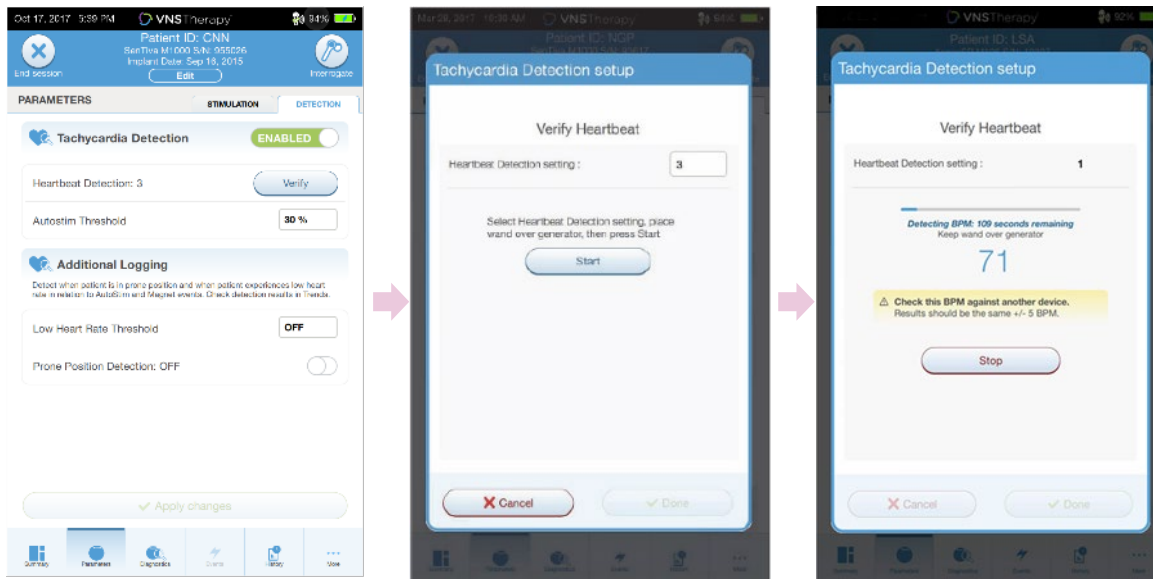
AspireSR™ (Model 106) & SenTiva™ (Model 1000)

Verify Heartbeat Detection

The Verify Heartbeat Detection verifies that the AutoStim-enabled device is correctly sensing the patient's heart rate

Performing the procedure step-by-step:

1. Interrogate device and temporarily turn off **ALL** output currents (program to 0 mA)
2. Go to **Detection** tab of the Parameters screen
3. Enable **Tachycardia Detection** (default Heartbeat Detection = 3, default Threshold for AutoStim = 70)
4. Select **Verify**



- “?? BPM” indicates lost/no communication, or no heart beats detected by the generator
- “<40 BPM” will display if the system detects a heart rate below this lower limit
- “>230 BPM “(Model 3000 v1.0) or “>180 BPM” (Model 3000 v1.5+) will display if the system detects a heart rate above this upper limit

Step 12: Irrigate and close the incision

- **Irrigate** both incision sites prior to closure
- To minimize scarring, a **subcuticular closure** is recommended for both skin incisions
- Administer **post-op antibiotics** (at physician discretion)





MRI Guidelines

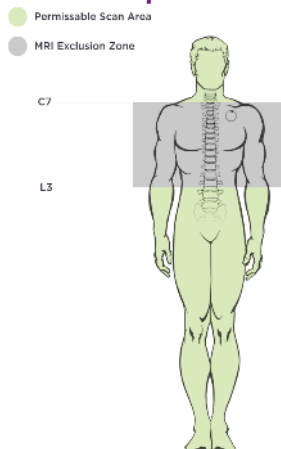
VNS Therapy™ and MRI

Performing MRI is safe, provided specified guidelines are followed*.

Scan Conditions: GROUP A

DemiPulse Model 103, AspireHC Model 105, AspireSR Model 106, SenTiva Model 1000 (generator in upper left chest, at or above armpit (above rib 4)†)

No special MRI equipment/coils required, a transmit body coil is permissible



Note: The scan iso-center must be outside the exclusion zone. Group B conditions provided on right side.

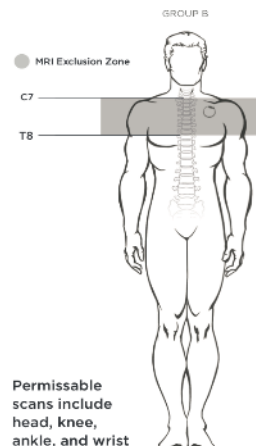
†Patients with implants in other locations must follow Group B scan conditions.

Scan Conditions: GROUP B

Applies to all Model 100, Model 101, Pulse Model 102, Pulse Duo™ Model 102R, DemiPulse Duo™ Model 104 (regardless of implant location) **AND** DemiPulse Model 103, AspireHC Model 105, AspireSR Model 106, SenTiva Model 1000 (in other implant locations**))

Requires local transmit-receive coil

Availability may vary



** implanted below armpit (lower than rib 4) or other implant locations

The device has been evaluated for MRI-induced risks, including heating, unintended stimulation, force, torque, device malfunction, and device vibration, and has been determined to be safe under the conditions specified in labeling; however, the patient may feel sensations of warmth or vibration at the implant site during the MRI scan

*Other scan conditions not specifically mentioned in this slide apply – see: <https://www.livanova.com/epilepsy-vnsthery/en-us/hcp/mri-safety>

MRI, magnetic resonance imaging. 1. VNS Therapy™ System Epilepsy Physician's Manual, May 2020, LivaNova USA, Inc. Houston, TX. Available at: <https://www.livanova.com/epilepsy-vnsthery/en-us/hcp/mri-safety>.

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Questions?

Additional information on SenTiva and programming can be found in the VNS Therapy™ System Epilepsy Physician's Manuals



Important Safety Information

Brief Summary¹ of Safety Information for the VNS Therapy™ System

[Epilepsy Indication] (February 2021)

1. INTENDED USE / INDICATIONS

Epilepsy (US)—The VNS Therapy System is indicated for use as an adjunctive therapy in reducing the frequency of seizures in patients 4 years of age and older with partial onset seizures that are refractory to antiepileptic medications.

2. CONTRAINDICATIONS

Vagotomy—The VNS Therapy System cannot be used in patients after a bilateral or left cervical vagotomy.

Diathermy—Do not use short-wave diathermy, microwave diathermy, or therapeutic ultrasound diathermy on patients implanted with a VNS Therapy System. Diagnostic ultrasound is not included in this contraindication.

3. WARNINGS — GENERAL

Physicians should inform patients about all potential risks and adverse events discussed in the physician's manuals. This document is not intended to serve as a substitute for the complete physician's manuals. The safety and efficacy of the VNS Therapy System have not been established for uses outside the "Intended Use/Indications" section of the physician's manuals.

The safety and effectiveness of the VNS Therapy System in patients with predisposed dysfunction of cardiac conduction systems (re-entry pathway) have not been established. Post-implant electrocardiograms and Holter monitoring are recommended if clinically indicated.

Postoperative bradycardia can occur among patients with certain underlying cardiac arrhythmias.

It is important to follow recommended implantation procedures and intraoperative product testing described in the *Implantation Procedure* chapter of the physician's manuals. During the intraoperative System Diagnostics (Lead Test), infrequent incidents of bradycardia and/or asystole have occurred. If asystole, severe bradycardia (heart rate < 40 bpm), or a clinically significant change in heart rate is encountered during a System Diagnostics

(Lead Test) or during initiation of stimulation, physicians should be prepared to follow guidelines consistent with Advanced Cardiac Life Support (ACLS).

Difficulty swallowing (dysphagia) may occur with active stimulation, and aspiration may result from the increased swallowing difficulties. Patients with pre-existing swallowing difficulties and those with a history of drooling or hypersalivation are at greater risk for aspiration. Use of the magnet to temporarily stop stimulation while eating may mitigate the risk of aspiration.

Dyspnea (shortness of breath) may occur with active VNS Therapy. Any patient with underlying pulmonary disease or insufficiency such as chronic obstructive pulmonary disease or asthma may be at increased risk for dyspnea.

Patients with obstructive sleep apnea (OSA) may have an increase in apneic events during stimulation. Lowering stimulus frequency or prolonging "OFF" time may prevent exacerbation of OSA. Vagus nerve stimulation may also cause new onset sleep apnea in patients who have not previously been diagnosed with this disorder.

Device malfunction could cause painful stimulation or direct current stimulation. Either event could cause nerve damage. Patients should be instructed to use the magnet to stop stimulation if they suspect a malfunction, and then to contact their physician immediately for further evaluation.

Patients with the VNS Therapy System, or any part of the VNS Therapy System, implanted should have MRI procedures performed only as described in the *MRI with the VNS Therapy System* instructions for use. In some cases, surgery will be required to remove the VNS Therapy System if a scan using a transmit RF body coil is needed.

Excessive stimulation at an excess duty cycle (that is, one that occurs when "ON" time is greater than "OFF" time) and high frequency stimulation (i.e., stimulation at ≥ 50 Hz) has resulted in degenerative nerve damage in laboratory animals.

Brief Summary¹ of Safety Information for the VNS Therapy™ System

[Epilepsy Indication] (February 2021)

Patients who manipulate the generator and lead through the skin (Twiddler's Syndrome) may damage or disconnect the lead from the generator and/or possibly cause damage to the vagus nerve.

The Wand, Programmer, and patient magnet are MR unsafe devices. These devices are projectile hazards and must not be brought into the MR scanner room.

Generators with AutoStim only—The AutoStim Mode feature should not be used in patients with clinically meaningful arrhythmias currently being managed by devices or treatments that interfere with normal intrinsic heart rate responses (e.g., pacemaker dependency, implantable defibrillator, beta adrenergic blocker medications). Patients also should not have a history of chronotropic incompetence [commonly seen in patients with sustained bradycardia (heart rate < 50 bpm)].

Generators with AutoStim only—For anticipated use of the AutoStim feature, it is important to follow the recommended pre-surgical surface assessment described in the Implantation Procedure to determine a location for the generator to reside in which it can accurately detect heart beats

4. WARNINGS — EPILEPSY

The VNS Therapy System should only be prescribed and monitored by physicians who have specific training and expertise in the management of seizures and the use of this device. It should only be implanted by physicians who are trained in surgery of the carotid sheath and have received specific training in the implantation of this device.

The VNS Therapy System is not curative. Physicians should warn patients that the VNS Therapy System is not a cure for epilepsy and that since seizures may occur unexpectedly, patients should consult with a physician before engaging in unsupervised activities, such as driving, swimming, and bathing, and in strenuous sports that could harm them or others.

Sudden unexpected death in epilepsy (SUDEP): Through August 1996, 10 sudden and

unexpected deaths (definite, probable, and possible) were recorded among the 1,000 patients implanted and treated with the VNS Therapy device. During this period, these patients had accumulated 2,017 patient-years of exposure. Some of these deaths could represent seizure-related deaths in which the seizure was not observed, at night, for example. This number represents an incidence of 5.0 definite, probable, and possible SUDEP deaths per 1,000 patient-years. Although this rate exceeds that expected in a healthy (nonepileptic) population matched for age and sex, it is within the range of estimates for epilepsy patients not receiving vagus nerve stimulation, ranging from 1.3 SUDEP deaths for the general population of patients with epilepsy, to 3.5 (for definite and probable) for a recently studied antiepileptic drug (AED) clinical trial population similar to the VNS Therapy System clinical cohort, to 9.3 for patients with medically intractable epilepsy who were epilepsy surgery candidates.

5. PRECAUTIONS — GENERAL

Physicians should inform patients about all potential risks and adverse events discussed in the VNS Therapy physician's manuals.

Prescribing physicians should be experienced in the diagnosis and treatment of epilepsy and should be familiar with the programming and use of the VNS Therapy System.

Physicians who implant the VNS Therapy System should be experienced performing surgery in the carotid sheath and should be trained in the surgical technique relating to implantation of the VNS Therapy System.

The safety and effectiveness of the VNS Therapy System have not been established for use during pregnancy. VNS should be used during pregnancy only if clearly needed.

The VNS Therapy System is indicated for use only in stimulating the left vagus nerve in the neck area inside the carotid sheath. The VNS Therapy System is indicated for use only in stimulating the **left vagus nerve below where the superior and inferior cervical cardiac branches separate from the vagus nerve**.

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[Epilepsy Indication] (February 2021)

It is important to follow infection control procedures. Infections related to any implanted device are difficult to treat and may require that the device be explanted. The patient should be given antibiotics preoperatively. The surgeon should ensure that all instruments are sterile prior to the procedure. Children 4-11 years of age may have a greater risk for infection when compared to adolescent and adult patients (≥ 12 years). Careful monitoring for site infection as well as the avoidance of manipulation of the surgical site post implant in children should be stressed.

The VNS Therapy System may affect the operation of other implanted devices, such as cardiac pacemakers and implanted defibrillators. Possible effects include sensing problems and inappropriate device responses. If the patient requires concurrent implantable pacemaker, defibrillatory therapy or other types of stimulators, careful programming of each system may be necessary to optimize the patient's benefit from each device.

Reversal of lead polarity has been associated with an increased chance of bradycardia in animal studies. It is important that the electrodes are attached to the left vagus nerve in the correct orientation. It is also important to make sure that leads with dual connector pins are correctly inserted (white marker band to + connection) into the generator's lead receptacles.

The patient can use a neck brace for the first week to help ensure proper lead stabilization.

Do not program the VNS Therapy System to an "ON" or periodic stimulation treatment for at least 14 days after the initial or replacement implantation.

For Models 100, 101, 102 and 102R do not use frequencies of 5 Hz or below for long-term stimulation.

Resetting the generator disables or turns the device OFF (output current = 0 mA). For Model 100, 101, 102 and 102R, resetting the generator will result in device history loss.

Patients who smoke may have an increased risk of laryngeal irritation.

Generators with AutoStim only—Because the device senses changes in heart rate, false positive detection may cause unintended stimulation. Examples of instances where the heart rate may increase include exercise, physical activity, and normal autonomic changes in heart rate, both awake and asleep, etc. Adjustments to the AutoStim feature's detection threshold should be considered; which may include turning the feature OFF.

Generators with AutoStim only—The physical location of the device critically affects the feature's ability to properly sense heart beats. Care must be taken to follow the implant location selection process outlined in the Implantation Procedure.

Generators with AutoStim only—Talk to your patient about use of the AutoStim feature since use of the feature will result in faster battery drain and the potential for more frequent device replacements. The physician's manual describes the impacts to the battery life. The patient should return to their physician at appropriate intervals to further evaluate whether they are receiving benefit from the current AutoStim settings.

M1000 only — Since the Scheduled Programming feature allows the generator to apply therapy increases at scheduled intervals, it may not be appropriate for use in patients who are nonverbal or are unable to use the patient magnet to stop undesired stimulation. Similarly, exercise caution for use of this feature in patients with a history of obstructive sleep apnea, shortness of breath, coughing, swallowing difficulties, or aspiration.

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[Epilepsy Indication] (February 2021)

6. ENVIRONMENTAL AND MEDICAL THERAPY HAZARDS

Patients should exercise reasonable caution in avoiding devices that generate a strong electric or magnetic field. If a generator ceases operation while in the presence of electromagnetic interference (EMI), moving away from the source may allow it to return to its normal mode of operation.

VNS Therapy System operation **should always be checked** by performing device diagnostics after any of the procedures mentioned in the physician's manuals.

For clear imaging, patients may need to be specially positioned for mammography procedures, because of the location of the generator in the chest.

Therapeutic radiation may damage the generator's circuitry, although no testing has been done to date and no definite information on radiation effects is available. Sources of such radiation include therapeutic radiation, cobalt machines, and linear accelerators. The radiation effect is cumulative, with the total dosage determining the extent of damage. The effects of exposure to such radiation can range from a temporary disturbance to permanent damage, and may not be detectable immediately.

External defibrillation may damage the generator.

Use of electrosurgery [electrocautery or radio frequency (RF) ablation devices] may damage the generator.

Magnetic resonance imaging (MRI) should not be performed using a transmit RF body coil for certain VNS therapy device configurations or under certain specific conditions. In some cases, heating of the lead caused by the transmit RF body coil during MRI may result in serious injury. Static, gradient, and radio frequency (RF) electromagnetic fields associated with MRI may change the generator settings (i.e., reset parameters) or activate the VNS device if the Magnet Mode output remains "ON". Note that certain magnetic resonance (MR) system head coils operate in receive-only mode and require use of the transmit RF body coil. Other MR systems use a transmit/receive RF head coil.

Local or surface coils may also be receive-only RF coils that require the transmit RF body coil for MRI. **The use of a receive RF coil does not alter hazards of the transmit RF body coil.** Exposure of the VNS Therapy System to any transmit RF coil must be avoided. Do not perform MRI scans using any transmit RF coil in the defined exclusion zones. See *MRI with the VNS Therapy System* instructions for use for details or further instructions for special cases such as lead breaks or partially explanted VNS Therapy systems.

Extracorporeal shockwave lithotripsy may damage the generator. If therapeutic ultrasound therapy is required, avoid positioning the area of the body where the generator is implanted in the water bath or in any other position that would expose it to ultrasound therapy. If that positioning cannot be avoided, program the generator output to 0 mA for the treatment, and then after therapy, reprogram the generator to the original parameters.

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If the patient receives medical treatment for which electric current is passed through the body (such as from a TENS unit), either the generator should be set to 0 mA or function of the generator should be monitored during initial stages of treatment.

Routine therapeutic ultrasound could damage the generator and may be inadvertently concentrated by the device, causing harm to the patient.

For complete information related to home occupational environments, cellular phones, other environmental hazards, other devices, and ECG monitors, refer to the physician's manuals.

7. ADVERSE EVENTS — EPILEPSY

Adverse events reported during clinical studies as statistically significant are listed below in alphabetical order: ataxia (loss of the ability to coordinate muscular movement); dyspepsia (indigestion); dyspnea (difficulty breathing, shortness of breath); hypoesthesia (impaired sense of touch); increased coughing; infection; insomnia (inability to sleep); laryngismus (throat, larynx spasms); nausea; pain; paresthesia (prickling of the skin); pharyngitis (inflammation of the pharynx, throat); voice alteration (hoarseness); vomiting.

¹The information contained in this Brief Summary for Physicians represents partial excerpts of important prescribing information taken from the physician's manuals. (Copies of VNS Therapy physician's and patient's manuals are posted at www.livanova.com.) The information is not intended to serve as a substitute for a complete and thorough understanding of the material presented in all of the physician's manuals for the VNS Therapy System and its component parts nor does this information represent full disclosure of all pertinent information concerning the use of this product, potential safety complications, or efficacy outcomes.

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See important safety information at:

<https://www.livanova.com/epilepsy-vnstherapy/en-us/safety-information>

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